Clara Rehmann

Ph.D. Candidate
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Education

Ph.D. Ecology and Evolutionary Biology — September 2021 - present
University of Oregon *Eugene, Oregon*
Anticipated graduation: April 2026

Bachelor of Science Biology — June 2020
University of Oregon *Eugene, Oregon* — September 2018 - June 2020
University of California, Los Angeles *Los Angeles, California* — September 2017 - June 2018

Research experience

PhD Student — September 2022 - present
Advisor: Dr. Andrew Kern, *Institute of Ecology & Evolution, University of Oregon*
Developing and applying geographically-informed population genetics methods with a focus on *Anopheles gambiae*. Projects include correcting for spatial bias in machine learning inference of location from genetic data, constructing open-source resources for population genetic simulation and analysis, and using novel space-based methods to identify positively-selected variants.

Rotation Student — March 2022 - June 2022
Advisor: Dr. William Cresko, *Institute of Ecology & Evolution, University of Oregon*
Assessed the impact of varying cross-species scRNA-seq integration methods on analysis outcomes in order to improve identification of developing facial structures in Syngnathids.

Rotation Student — September 2021 - December 2021
Advisor: Dr. Michael Harms, *Institute of Molecular Biology, University of Oregon*
Investigated population-level effects on possible evolutionary trajectories across a fitness landscape with high-order epistasis using individual-based simulation.

Scientific Programmer — June 2020 - June 2021
Advisor: Dr. Andrew Kern, *Institute of Ecology & Evolution, University of Oregon*
Developed Python plotting tools for machine learning methods and assessed the impact of biased dispersal on machine learning inference of location from genetic data.

Research Assistant — April 2020 - June 2021
Advisors: Dr. Matthew Barber, Dr. Kristin Kohler, *Institute of Ecology & Evolution, University of Oregon*
Used computational and experimental methods to investigate coevolutionary dynamics between primate lysozyme and bacterial lysozyme inhibitors.

Undergraduate Researcher — September 2017 - June 2018
Advisor: Dr. Beth Lazazzera, *University of California, Los Angeles*
Experimental work investigating impacts of inaccurate translation caused by the loss of aminoacyl-tRNA synthetase editing on *E. coli* fitness.

Teaching Experience

Evolutionary Genetics Journal Club Fall 2024 - present
Instructor *Department of Biology, University of Oregon*
Organized and facilitated weekly meetings about current publications.

Evolutionary Biology Fall 2023
Guest Lecturer *Department of Biology, University of Oregon*
Lectured on integrating evolutionary forces.

Computational Genetics Spring 2023
Guest Lecturer *Department of Biology, University of Oregon*
Lectured on principal component analysis and data visualization.

Evolutionary Biology Fall 2022
Guest Lecturer *Department of Biology, University of Oregon*
Lectured on genotypic and phenotypic variation.

General Biology IV: Mechanisms Spring 2022
Graduate Employee *Department of Biology, University of Oregon*
Lead and instructed students during wet and dry lab activities, graded assignments and assisted in curriculum development.

Immunology and Infectious Disease Winter 2022
Graduate Employee *Department of Biology, University of Oregon*
Developed discussion material, lead and instructed students during discussions, and graded assignments.

Evolutionary Biology Fall 2021
Graduate Employee *Department of Biology, University of Oregon*
Lead student discussions of publications in evolutionary biology, and graded assignments.

Marine Biology Winter 2020
Biology Undergraduate Lab Assistant *Department of Biology, University of Oregon*
Assisted in instruction of students during wet lab activities.

General Biology IV: Mechanisms Summer 2019
Biology Tutor for Undergraduates *Department of Biology, University of Oregon*
Assisted in instruction of students during lectures and wet and dry lab activities

Publications

Gower G, Pope NS, Rodrigues MF, Tittes S, Tran LN, Alam O, Cavassim MIA, Fields PD, Haller BC, Huang X, Jeffrey B, Korfmann K, Kyriazis CC, Min J, Rebello I, **Rehmann CT**, Small ST, Smith CCR, Tsambos G, Wong Y, Zhang Y, Huber CD, Gorjanc G, Ragsdale AP, Gronau I, Gutenkunst RN, Kelleher J, Lohmueller KE, Schrider DR, Ralph PL, Kern AD. Accessible, realistic genome simulation with selection using stdpopsim. Molecular Biology and Evolution, msaf236, September 2025. DOI:10.1093/molbev/msaf236

Rehmann CT, Small ST, Ralph PL, Kern AD. Sweeps in space: Leveraging geographic data to identify beneficial alleles in *Anopheles gambiae*. Molecular Biology and Evolution, Volume 42, Issue 6, June 2025, msaf141. DOI: 10.1093/molbev/msaf141

Chevy E, Miseon J, Caudill V, Champer S, Haller B, **Rehmann CT**, Smith CS, Tittes S, Messer P, Kern AD, Ramachandran S, Ralph PL. Population genetics meets ecology: a guide to individual-based simulations in continuous landscapes. Ecology and Evolution, Volume 15, Issue 4, March 2025, DOI: 10.1002/ece3.71098

Rehmann CT, Ralph PL, Kern AD. Evaluating evidence for co-geography in the *Anopheles Plasmodium* host-parasite system. G3 Genes—Genomes—Genetics, Volume 14, Issue 3, March 2024, DOI: 10.1093/g3journal/jkae008

Presentations

ESPOD Fellowship Talks December 2025

European Bioinformatics Institute, *Hinxton, United Kingdom*

Invited talk: Bugs In Space: Spatial modeling of *Anopheles* population genetic dynamics for vector control

Matheison Lab December 2025

University of Pennsylvania, *Philadelphia, Pennsylvania, USA*

Invited talk: Bugs In Space: Understanding *Anopheles* and *Plasmodium* evolution with spatial population genetics

Goldberg Lab December 2025

University of California, Los Angeles, *Los Angeles, California, USA*

Invited talk: Bugs In Space: Understanding *Anopheles* and *Plasmodium* evolution with spatial population genetics

Biology of Genomes May 2025

Cold Spring Harbor Laboratory, *Cold Spring Harbor, New York, USA*

Poster: In space no one can hear you sweep: Leveraging geographic data to identify beneficial alleles in *Anopheles gambiae*

EVO-WIBO April 2025

American Society of Naturalists, *Blaine, Washington, USA*

Poster: In space no one can hear you sweep: Leveraging geographic data to identify beneficial alleles in *Anopheles gambiae*

Malaria Genomics Journal Club 24 September 2024

Malaria Genomic Epidemiology Network, *virtual*

Invited talk: Location, location, location: Evaluating evidence for co-geography in the Anopheles Plasmodium host-parasite system

Genetic Epidemiology of Malaria September 2024

Wellcome Sanger Institute, *Hinxton, United Kingdom*

Poster: In space no one can hear you sweep: Dynamics and outcomes of selective sweeps in continuous geographic space

SMBE July 2024

Society for Moleculary Biology and Evolution, *Puerto Vallarta, Jalisco, Mexico*

Talk: In space no one can hear you sweep: A novel signal of selective sweeps in continuous geographic space

The Allied Genomics Conference March 2024

Genetics Society of America, *National Harbor, Maryland, USA*

Poster: In space no one can hear you sweep: Dynamics and outcomes of selective sweeps in continuous geographic space

Lighthouse Lightning Talks 2 February 2024
Verena Viral Emergence Research Institute, *virtual*
Invited talk: Location, location, location: Evaluating evidence for co-geography in the Anopheles Plasmodium host-parasite system

SMBE July 2023
Society for Moleculary Biology and Evolution, *Ferrara, Italy*
Poster: Location, location, location: Evaluating evidence for co-geography in the Anopheles Plasmodium host-parasite system

EVO-WIBO April 2023
American Society of Naturalists, *Port Townsend, Washington, USA*
Poster: Location, location, location: Dissecting errors in machine learning prediction of geography

Population, Evolutionary, and Quantitative Genetics June 2022
Genetics Society of America, *Asilomar, California, USA*
Poster: Location, location, location: Dissecting errors in machine learning prediction of geography

Probabilistic Modeling in Genomics April 2021
Cold Spring Harbor Laboratories, *virtual*
Poster: Location, location, location: Dissecting errors in machine learning prediction of geography

Outreach and Community Involvement

Women in Graduate Science Fall 2021 - present

Webmaster Academic Year 2025 - 2026
Updating the WGS website and social media pages; promoting WGS events and sharing information from associated organizations.

President Academic Year 2024 - 2025
Conducting and aiding in organization of WGS meetings and events; collaborating with the Graduate School, UO Foundation, and Associated Students of the UO; promoting the mission of WGS.

Public Relations Chair Academic Year 2023 - 2024
Communicating with WGS members; advertising WGS events to the on- and off-campus community.

Social Co-chair Academic Year 2022 - 2023
Planning and organizing social events for WGS members and the local community.

Mad Duck Outreach Winter 2023 - present
Helping facilitate on-campus science activities for middle school students.

Graduate Evolutionary Biology and Ecology Students Fall 2021 - present

Public Relations & Website Organizer Academic Year 2025 - 2026
Advertising GrEBES events to the on- and off-campus community; designing and maintaining organization website.

SPECIE2S chair Academic year 2023 - 2024
Organizing and facilitating weekly writing groups and peer feedback sessions.

Seminar committee Academic Year 2022 - 2023; 2023 - 2024; 2024 - 2025

Helping plan and organize an annual free public seminar on current topics in biology.

Genetics Society of America Member Winter 2022 - present

American Society of Naturalists Winter 2023 - present

Society for Molecular Biology and Evolution Spring 2023 - present